Species Narratives

Species Narratives summarize the sensitivity, exposure, and vulnerability results for each species, and provide additional context on known climate change impacts on the species and known life history characteristics.

PDF versions of the narratives will be available for download.

All figures can be enlarged by clicking on them.

Each narrative contains the information described below. Methods are briefly described for all metrics. Please see the publication for additional details.

Summary

A summary section providing the final average Sensitivity, Exposure, and Vulnerability scores. If scores were calculated across species stocks and scores differed across stocks, a range of scores will be provided.

This section also contains a summary table containing the final average scores and the distribution of expert tallies (Sensitivity) or total scores across space (Exposure/Vulnerability).

Changes from Previous Assessments

This section contains a table highlighting the scores from this assessment, and compares them to the previous assessment(s). If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Stocks

If a species was scored within different stock groups, a map of the different stocks is provided here and referenced throughout the narrative. If the species is not scored within different stock groups, no plot is provided.

Score Descriptions

These sections go into the scores in-depth, describing average scores, spatial patterns, and drivers.

Sensitivity

Sensitivity scores were estimated using expert solicitation following Hare et al. (2016). A subset of sensitivity attributes were updated from the original analysis to reflect new insights of climate change on species life histories. Life history data quality was ranked by experts using a tally system to help quantify certainty. Sensitivity certainty was calculated by applying bootstrapping methods to expert tallies as in Hare et al. (2016).

Exposure

Exposure scores were estimated in a two steps. First, raw exposure was calculated monthly across the Northeast Shelf for each exposure factor used in the species distribution models and ranked following the original analysis Hare et al. (2016) and outlined in the table below.

Table 1: Raw exposure scores and corresponding ranks

Score	Raw Exposure Score	Rank
Very High	> 2	4
High	1.5 - 2	3
Moderate	0.5-1.5	2
Low	<0.5	1

Second, monthly exposure ranks for each factor were averaged using the species distribution model results as weights to account for where and when species were present. Averaging occurred as across months to produce a map of mean exposure across space.

In this version of the analysis, the present day time period was defined as **2014-2023** and the future time period was defined as **2025-2035**. For reference, Hare et al. (2016) used 1956-2025 and 2006-2055 as their present and future timeframes, respectively.

Creating Total Scores and Vulnerability

Total sensitivity and exposure scores were calculated using the same logic rule as Hare et al. (2016):

Table 2: Logic rule from Hare et al. (2016) used to calculate total sensitivity and exposure scores

Overall Sensitivity or Exposure Score	Numeric Score	Logic Rule	
Very High	4	3 or more attributes or factors	3.5
High	3	2 or more attributes or factors	3.0
Moderate	2	2 or more attributes or factors	2.5
Low	1	All other scores	

For exposure scores, total scores were calculated spatially, meaning that the logic rule was applied to each grid cell.

Vulnerability was calculated by multiplying the numeric exposure and sensitivity scores following Hare et al. (2016). In this analysis, this calculation was done spatially (within each map grid cell).

Total Scores Across Space

Vulnerability and exposure scores were calculated by averaging results across the target species' range and within defined stocks if available. Average overall scores were defined as the following:

Table 3: Categorical scores corresponding to average vulnerability and exposure scores

Average Vulnerability or Exposure Score	Average Exposure Numeric Score	Average Vulnerability Numeric Score
Very High	≥ 3.5	≥ 10.5
High	< 3.5	< 10.5
Moderate	< 2.5	< 7.0
Low	< 1.5	< 3.5

Data Quality Metrics

This section describes the Life History Data Quality and Model Confidence Scores. These scores quantify how confident experts were in the data provided that were used to generate the sensitivity (Life History Data Quality) and exposure (Model Confidence) scores. Model forecast data was not scored, as it was in the previous assessment, because forecast data received high data quality scores in the previous assessment. Both data quality scores were provided by the experts and range from 0 - 3. Following the original analysis, a score above 2 was deemed acceptable (Hare et al. 2016).

Life History Data Quality

Life History Data Quality replaces the Data Quality score for the sensitivity attributes. It was calculated using identical methods to the original analysis (average of expert-provided scores) and uses the same rubric (Hare et al. 2016).

Table 4: Rubric used by experts to score Life History Data Quality

Life History Data Quality Score	Description
3	**Adequate Data.** The score is based on data which have been observed, modeled or empirically measured for the species in question and comes a from reputable source.
2	**Limited Data.** The score is based on data which has a higher degree of uncertainty. The data used to score the attribute may be based on related or similar species, come from outside the study area, or the reliability of the source may be limited.
1	**Expert Judgment.** The attribute score reflects the expert judgment of the reviewer and is based on their general knowledge of the species, or other related species, and their relative role in the ecosystem.
0	**No Data.** No information to base an attribute score on. Very little is known about the species or related species and there is no basis for forming an expert opinion (please use judiciously).

Model Confidence Scores

The confidence in the species distribution models used to quantify exposure was evaluated both analytically and with expert feedback.

Area under the Curve

Model confidence was calculated analytically using the Area under the Receiver-Operator Curve (AUC). This statistic quantifies how well models classify binary data. The models were built on MOM6 hindcast data from 1993-2019 and AUC scores were calculated on presence/absence data and model predictions from 2020-2023 to explicitly test how well the models predict future conditions, especially in data-poor years such as 2020. The closer the AUC is to one, the better the model did at predicting observed presences and absences.

It is important to note that the AUC is limited by the scope of the data used to calculate it (in this case, the spatial/temporal distribution of samples from 2020-2023). For example, the AUC may not reflect poor predictions of a species presence outside of its known range if the region is undersampled (Type I error). Therefore, we enlisted the experts to provide the additional Model Confidence Score.

Model Confidence Score

The model confidence score was modeled off of the original Data Quality metric (now called Life History Data Quality) (Hare et al. 2016). This score asked the experts to evaluate whether the final species distribution ensemble model results met their expectations based on their knowledge of the species and the life history data provided using the rubric below:

Table 5: Rubric used by experts to score Model Confidence

Model Confidence Score	Description
3	**High Confidence.** The model highly accurately reflects the species' realized niche. It captures fine-scale nuances, matches expert field observations, and handles environmental transitions realistically.
2	**Moderate Confidence.** The model aligns well with known distributions and environmental drivers. Most key habitats are represented, though there may be minor inaccuracies at the range edges or in data-poor regions.
1	**Low Confidence.** The model captures broad trends but has significant "noise." Environmental drivers likely don't reflect actual species needs.
0	**No Confidence.** The model fails to represent known species range. It predicts species in unsuitable areas or misses well-known core areas entirely.

This rubric was developed to mimic the Life History Data Quality rubric.

Additional Metrics

Distribution Shifts

Distribution Shifts were previously assessed using a subset of Sensitivity Attributes (Hare et al. 2016). In this analysis, they are quantified with the ensemble species distribution models.

Three metrics are calculated to help describe how species distributions shift:

1. **Weighted centroid**: the mean location of the species distribution. Species distribution model results are used as weights in the averaging to help account for habitat use. This is similar to the Center of Gravity metric used by DisMAP.

2. **Core Habitat & Range Area**: the area defined as core habitat and species range. Core habitat and range are quantified 50 and 95% Kernel Density Estimator (KDE), respectively.
3. **Leading and Trailing Edges**: the 95th and 5th percentiles of latitude (north-south) and longitude (east-west) edges of the species distribution. The difference between the leading and trailing edges, representing the span of the species distribution in degrees, is presented as the latitude/longitude range.

The provided map illustrates average species ranges using 5 year averages from the first and last 5 years of the hindcast data and the last 5 years of the most recent decadal forecast. The provided timeseries illustrate how a species' core habitat and range areas change throughout the MOM6 hindcast and decadal forecasts.

In the online version of the narratives, an animated gif is also provided to illustrate how species distributions and the leading and trailing edges of these distributions shift through time in both the hindcast and forecasts.

Directionality

Directional Effect is consistent with the previous analysis: experts were asked to score how they left the species would be impacted (positive/negative/neutral) by climate change (Hare et al. 2016). Scores were provided using a tally system, and certainty was calculated using the bootstrapping methods described in Hare et al. (2016), similar to the sensitivity analysis.

Climate Effects & Life History Synopsis

These sections of the narratives include summaries of known climate impacts on a species and a summary of their life histories from the literature.

Supplemental

Additional metrics and figures are provided in the supplement. The supplement currently includes the following sections:

Exposure Timeseries

Ranked exposures were calculated monthly across space. In the analysis described above, scores are averaged across months, using the species distribution model results as weights, to create a single average exposure map for each factor. Scores were also averaged across space to create a single score for each month. Averaging occurred over the species' range and within stock units if applicable. Total exposure was then calculated across time using the Hare et al. (2016) logic rule.

Timeseries for each exposure factor and total exposure are provided in this section.

Exposure due to Significant Factors

The methods described above assume that all of the possible exposure factors will impact the species fitness/vulnerability in the future. However, this may not be the case. Therefore, we calculated total exposure and total vulnerability using only the environmental factors that played a significant role in the final species distribution model.

Variable importance was calculated using model-specific methods, scaled between 0 and 1 for each component model, and averaged across models according to component model performance. Static variables such as bathymetry, rugosity, and distance to the coast were excluded. An exposure factor was considered important to the species distribution if its average importance was greater than 0.1 (indicating that the factor explains at least 10% of the variance explained by dynamic variables in the final ensemble distribution model). A radar plot is provided illustrating the factor importance in the model.

This section describes the total exposure results across time and space when only the factors that play a significant role in the species distribution model are considered. Both maps and timeseries are presented.

References

Hare, Jonathan A, Wendy E Morrison, Mark W Nelson, Megan M Stachura, Eric J Teeters, Roger B Griffis, Michael A Alexander, et al. 2016. "A Vulnerability Assessment of Fish and Invertebrates to Climate Change on the Northeast US Continental Shelf." *PloS One* 11 (2): e0146756.